

RSC Class 4

Market Competition: Monopoly and Perfect Competition

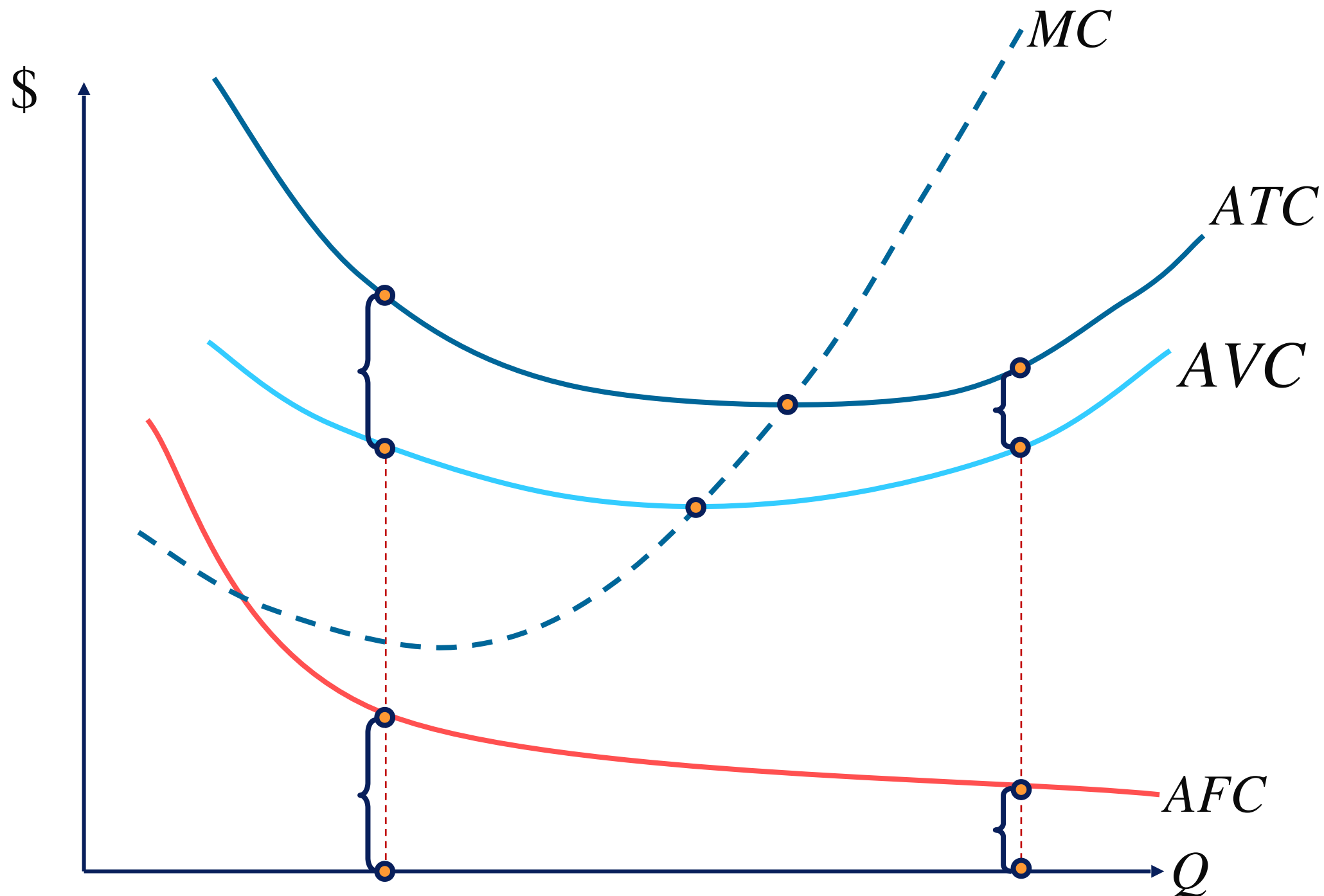
Meng Hsuan “Rex” Hsieh

Agenda for Today

- Costs
- Monopoly
- Perfect competition
- Firm's supply curve in perfectly competitive markets
- Short-run versus long-run equilibria in perfectly competitive markets
- ... Quiz Questions for Next Class!

Costs

Where we left off... we're now talking about firms



Costs

We're now talking about firms

What costs *matter* for firms' decision-making process?

Remember that firms maximize profits by setting the quantity according to the fundamental decision rule, which is:

_____ equal _____

Thinking on the margin $\Rightarrow$ any cost incurred in the past does not matter

Costs

We're now talking about firms

What costs *matter* for firms' decision-making process?

Def. Costs that occurred in the past, and are irrecoverable, are **sunk costs**.

Sunk costs do not matter from the point of view of decision-making

Thinking that they do is called sunk cost fallacy

Costs

We're now talking about firms

What costs *matter* for firms' decision-making process?

Example. Rex is a licensed headphone reseller and owns a shop. He bought 1,000 units of the latest Sony wireless headphones for \$500/unit.

Every pair of headphone purchased requires an initial maintenance, costing \$160/unit per year. This is an unavoidable cost.

Right now, he received an offer from another reseller, who offers to buy all 1,000 units from him for \$450/unit. Should he sell all of his stock or not?

Costs

We're now talking about firms

What costs *matter* for firms' decision-making process?

Example. Rex is a licensed headphone reseller and owns a shop. He bought 1,000 units of the latest Sony wireless headphones for \$500/unit.

To run the shop, he hires 18 clerks at \$55,000/year; pays \$30,000 in utilities per year; pays himself \$270,000/year.

Every pair of headphone purchased requires an initial maintenance, costing \$160/unit per year. This is an unavoidable cost.

He sold all 1,000 headphones. He's got a buyer for another headphone (of the same model). What is his marginal cost of selling 1 additional unit of the same headphone model?

Costs

We're now talking about firms

We've nailed down consumer behavior, firm's marginal revenue, and firm's costs — separately

We are now ready to put all of these together and look at a market, with supply and demand, together

The simplest form of competition: there is none!

This is called monopoly

Monopoly

Markets and Competition

We left off here...

BUSINESS | MEDIA

Judge Rules Google Operates Illegal Ad Monopoly

Tech giant faces multiple legal threats related to how it wields market power

By [Jan Wolfe](#) [Follow](#) and [Patience Haggin](#) [Follow](#)

Updated April 17, 2025 at 9:10 pm ET



128



Gift unlocked article

The Federal Trade Commission this week went to trial in [its antitrust case](#) against [Facebook](#) parent [Meta Platforms](#), with Chief Executive [Mark Zuckerberg](#) testifying on the stand for several days. The FTC is seeking to break up Meta's business, forcing the company to separate from Instagram and WhatsApp.

The FTC has accused [Amazon](#) of creating an illegal monopoly in the e-commerce space. A judge last year granted part of Amazon's request to dismiss the case early, and the company has said the suit is "wrong on the facts and the law."

[Apple](#) faces a [Justice Department antitrust case](#) over allegations it impedes the ability of outside software to integrate with its devices, locking users into its ecosystem. Apple has sought dismissal of the case, arguing the government has failed to show harm to smartphone users.

In August, U.S. District Judge Amit Mehta of Washington, D.C., ruled that [Google's dominant search engine is an unlawful monopoly](#).

Markets and Competition

We left off here...

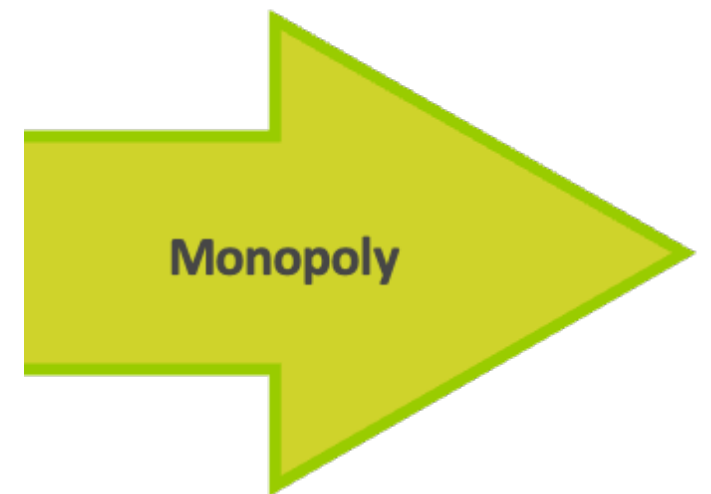
Let us focus on the case where there is no competition

In other words:

The firm captures all of market demand

⇒ We can use all the tools we have already developed to understand how monopolists think about consumers!

Only one firm
Full market power



Markets and Competition

Monopoly

There is now only one firm capturing the entire market demand for a product

In other words, market demand = demand that the monopolist faces

What is the fundamental decision rule for profit-maximizing firms? And what does this rule decide — quantity, or price?

Markets and Competition

Monopoly

What is the fundamental decision rule for profit-maximizing firms?
And what does this rule decide — quantity, or price?

Given a profit-maximizing quantity, what price should the monopolist sell at, and why?

Markets and Competition

Monopoly

Example. A monopolist faces demand for their product of $Q_D = 14 - 3P_D$. They have marginal cost function $MC(Q) = 2$. Determine their profit-maximizing quantity (to produce), price, and their profits at these choices.

Markets and Competition

Monopoly

Example, Redux. A monopolist faces demand for their product of $P_D = 10 - 2Q_D$. They have marginal cost function $MC(Q) = 2Q$. Determine their profit-maximizing quantity (to produce), price, and their profits at these choices.

Perfectly Competitive Markets

Markets and Competition

We leave the world of there being no competitor(s)

A natural question is: **what changes to a firm's decisions when there are competitors?**

Turns out answering this requires

1. Knowing how many competitors there are, and
2. Thinking about the implication of point 1 for equilibrium

Let's start with point 1

Markets and Competition

Perfect competition

1. Knowing how many competitors there are

The other extreme view of market reality is one where there are

- Large numbers of sellers and buyers; and
- Only identical products sold in the market, with no product differentiation

These imply sellers and buyers have no influence over the market price — the (aggregate) demand and supply decides prices!

This is the setting of **perfect competition**

Markets and Competition

Perfect competition

1. Knowing how many competitors there are

Def. Perfect competition is a form of competitive environment characterized by large number of sellers and buyers, with a homogeneous, undifferentiated product, where every firm and every individual takes *market prices as a given*.

Examples. Markets of produce (corn, rice, grains...). Markets of knockoffs.

Markets and Competition

Perfect competition

Few real-life examples come close to perfectly competitive markets, but some are somewhat close (though not always naturally)

China auto market price war stokes fears of industry shake-out

By Norihiko Shirouzu

May 27, 2025 4:37 PM EDT · Updated May 27, 2025



Of the 169 automakers operating in China today, more than half have less than 0.1% market share, according to data from research firm Jato Dynamics. The crowded field is reminiscent of the U.S. auto sector in the early 20th century, when more than 100 companies vied with big players such as Ford, before the industry consolidated.

Markets and Competition

Perfect competition

1. Knowing how many competitors there are

Implications of perfect competition. There are a few consequences:

- A. Market prices are determined by market supply and market demand.
- B. Firms cannot set their own price; they take the market price as given, implying the demand they face is _____.
- C. Firms maximize profits by setting the quantity according to the fundamental decision rule, which is:

_____ equal _____

Markets and Competition

Perfect competition

1. Knowing how many competitors there are

Question. What does A. and B. imply about the marginal revenue of a firm in a perfectly competitive market?

- A. Market prices are determined by market supply and market demand.
- B. Firms cannot set their own price; they take the market price as given.

This is an important fact to remember!

Markets and Competition

Perfect competition

1. Knowing how many competitors there are

Question. What does A., B., and C. imply about the market demand versus the demand the firm faces?

- A. Market prices are determined by market supply and market demand.
- B. Firms cannot set their own price; they take the market price as given.
- C. Firms maximize profits by setting the quantity according to the fundamental decision rule

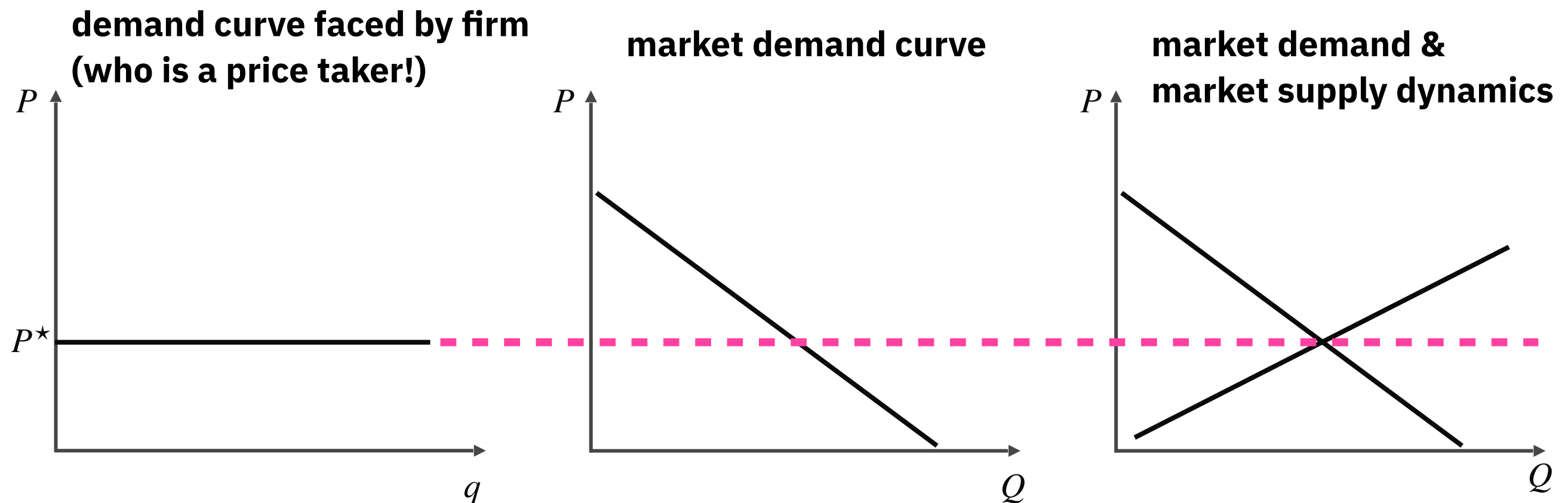
This is another important fact to remember!

Markets and Competition

Perfect competition

1. Knowing how many competitors there are

Putting everything together:



Markets and Competition

Perfect competition

1. Knowing how many competitors there are

Therefore, perfectly competitive markets are characterized by

- Large numbers of sellers and buyer + only identical products sold in the market, with no product differentiation
- Market prices are taken as given by everyone
- Demand each firm faces is perfectly elastic
- Each firm's supply is determined by its marginal costs (with some additional twists — we will start with this next time!)

Markets and Competition

Perfect competition

2. Thinking about the implication of point 1 for equilibrium

So far, we described nothing about how firms and individuals would behave in equilibrium in a perfectly competitive market

The fact that every firm takes prices as given — rather than setting prices themselves — must have consequences for the market equilibrium!

Markets and Competition

Perfect competition

Firms make decisions on the margin, so they rely on marginal cost to decide their profit-maximizing production quantity

We may naturally think their marginal cost function describes their supply decision

This turns out to not be quite right

Nuance comes from what types of costs are considered in our definition of market equilibrium

Markets and Competition

Perfect competition

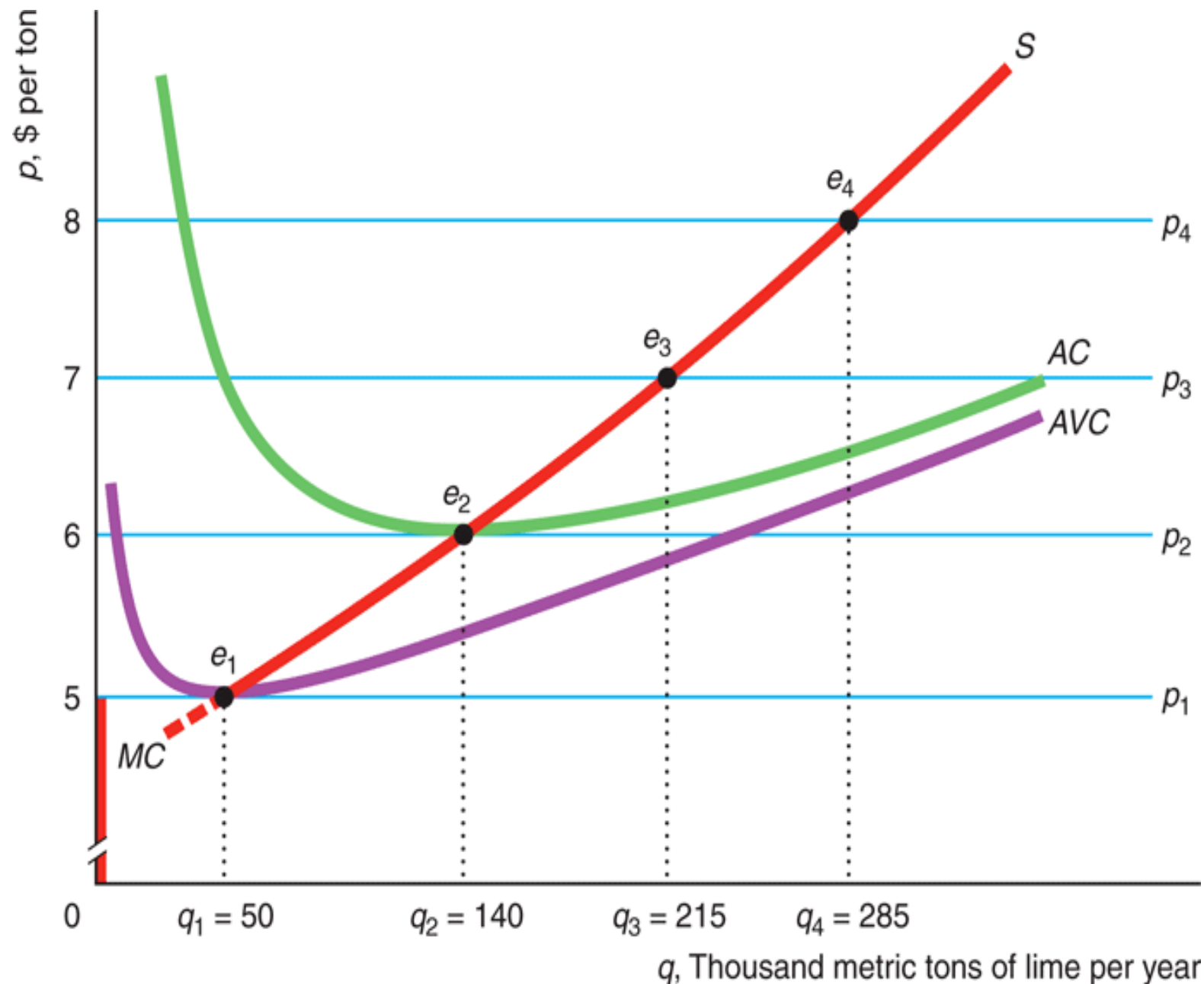
Market equilibrium is a statement about at what price and quantity the market demand “equal” market supply

Every firm in perfectly competitive market takes the market price as given, so this definition of market equilibrium has no “bite” for whether a firm would make positive economic profits

To determine profits, we need to look at the firm’s cost functions

Markets and Competition

Perfect competition



Firms produce quantity at which MC intersects $MR = \text{market price}$

Firm's MC traces out quantity it produces at each market price P

Economic profits at $e_2 = \$$ _____.

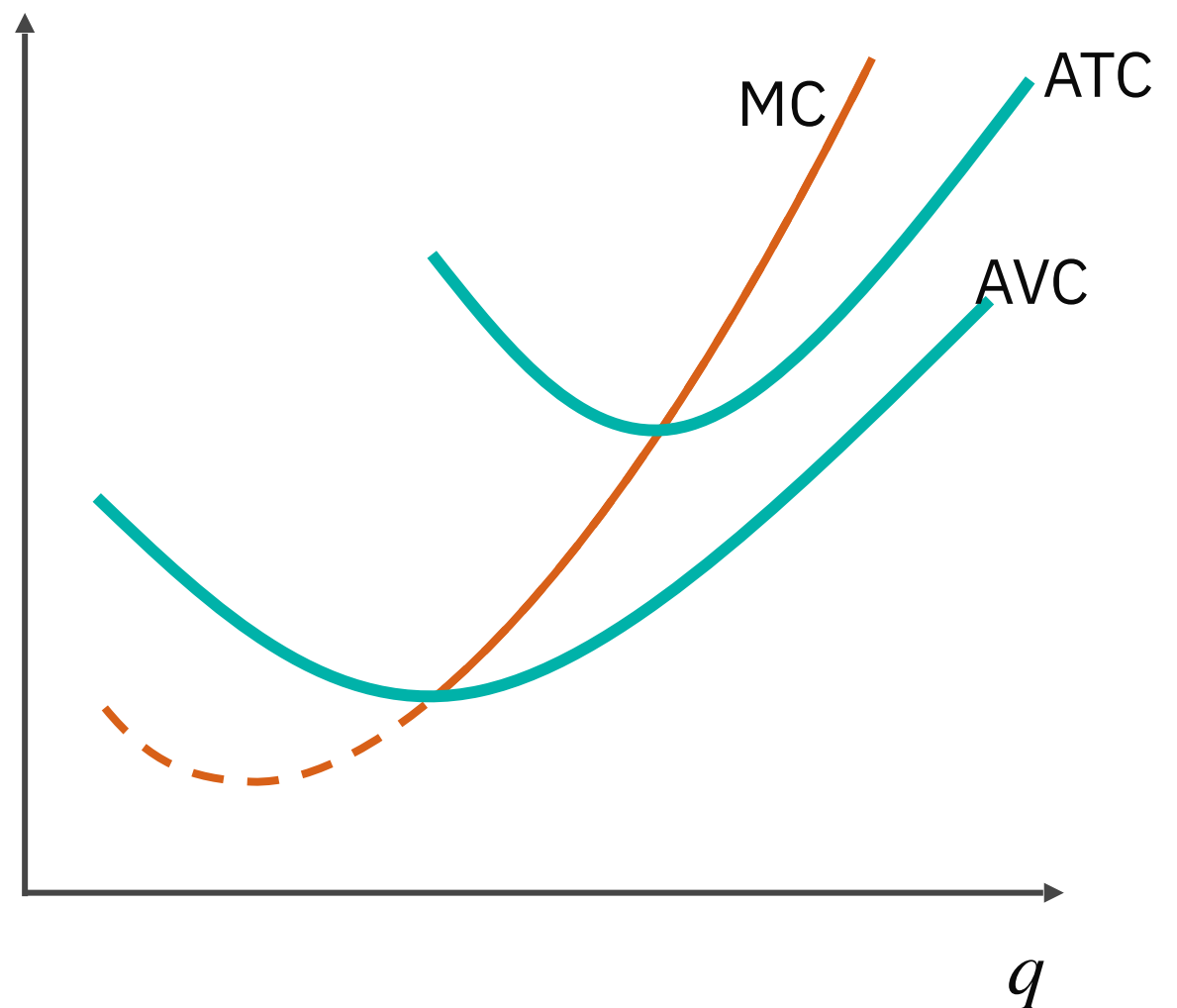
For market price $< AVC$, i.e. below e_1 , economic profits is _____ \$_____ $\Rightarrow$ firm shuts down!

Markets and Competition

Perfect competition

The analysis in the previous slide P is for when the market prices come from a *short-run (perfectly competitive) equilibrium*

Def. Short-run equilibrium is where firms consider both fixed and variable costs in making their production decisions



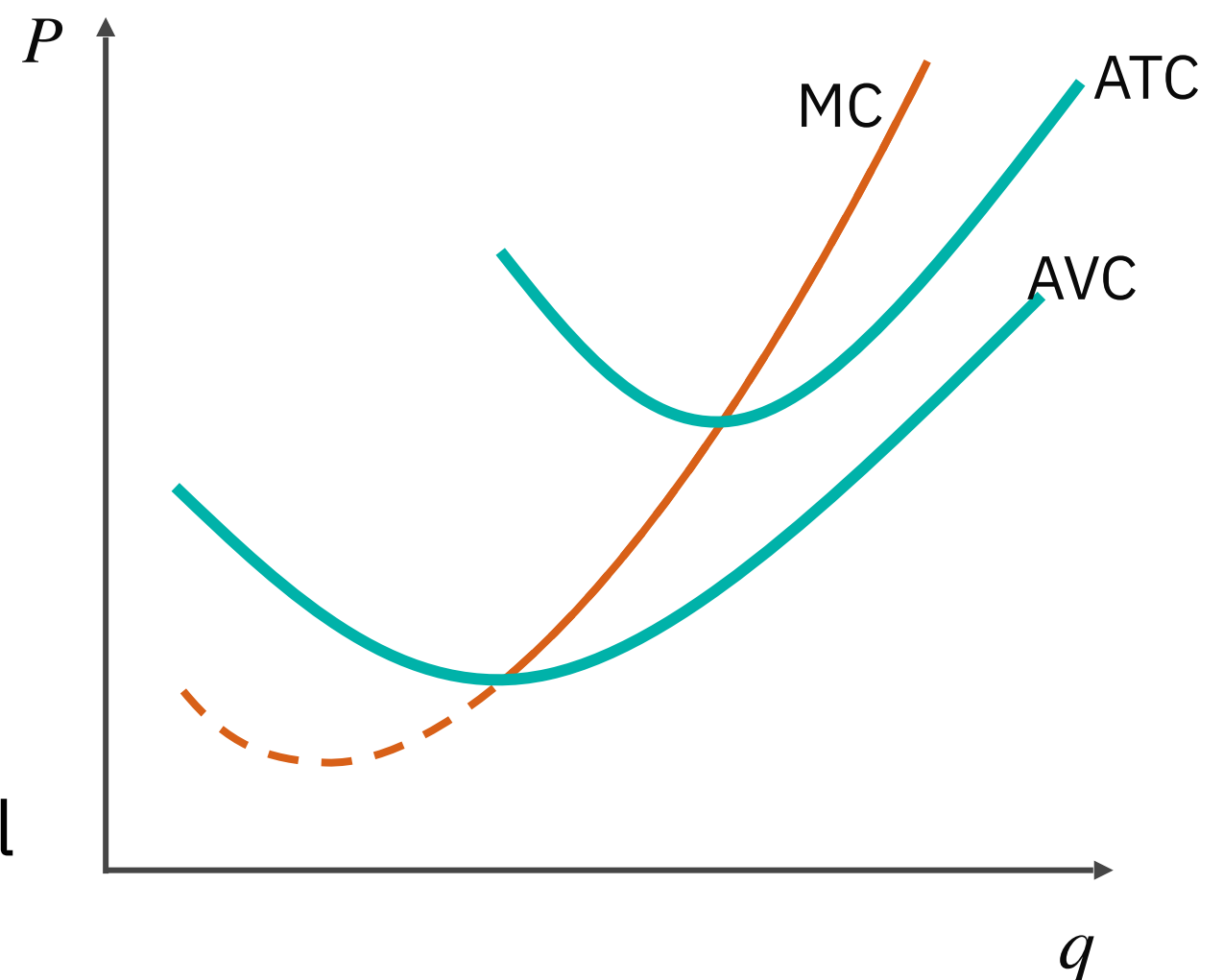
Markets and Competition

Perfect competition

Short-run supply curve of individual firm in PC market is section of MC curve above AVC

Firms shut down in short run when price $<$ AVC (operating profits are negative)

Takeaway: in the short-run market equilibrium, firms will still operate even if they make operating loss; if they cannot cover their variable costs, they will shut down



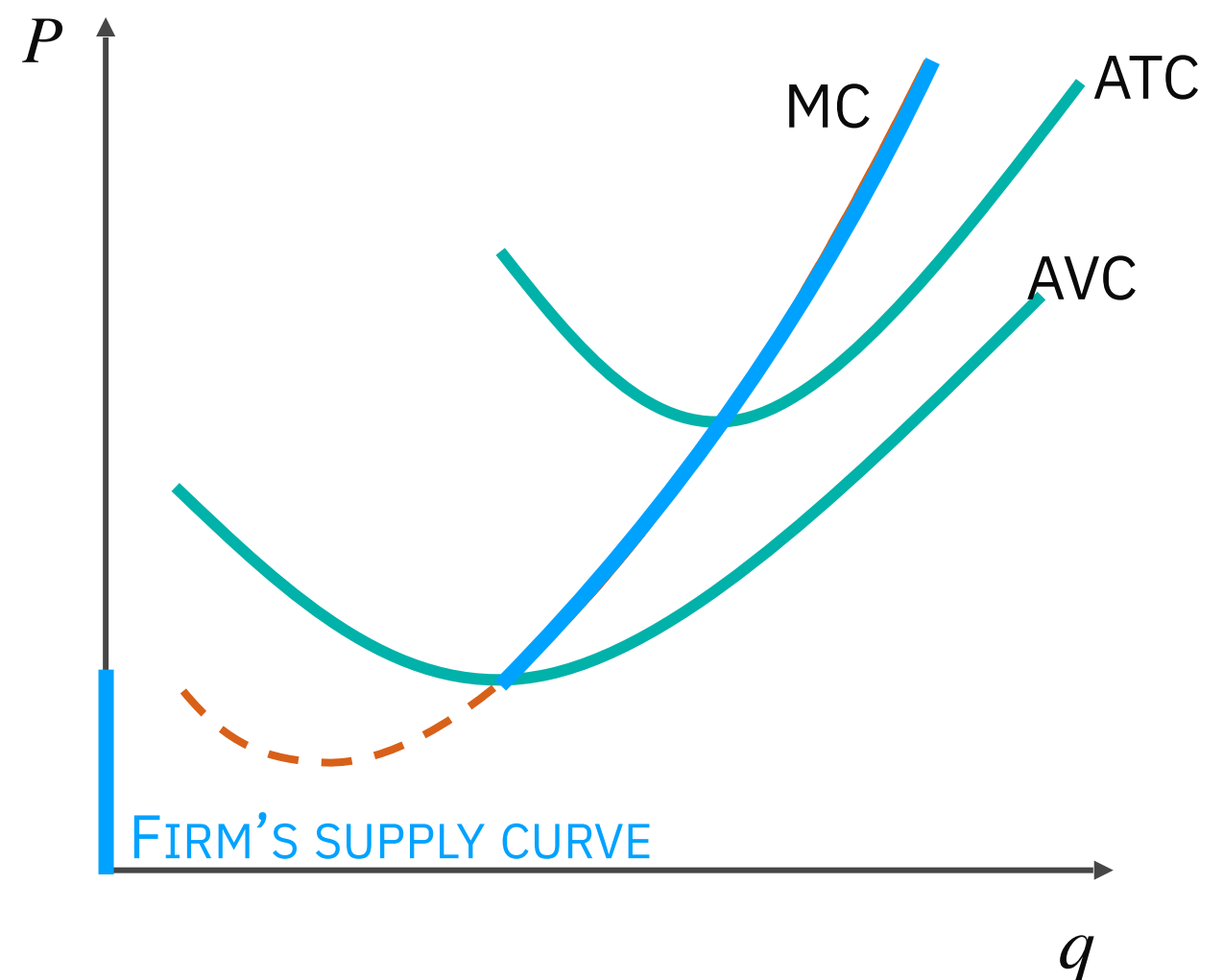
Markets and Competition

Perfect competition

Short-run is also characterized by free entry and exit due to no barriers to entry

Every firm is selling identical, undifferentiated products, so it is completely reasonable to assume barriers to entry is low

Why might new firms enter the market, and existing firms exit, if everyone takes the market price as given?



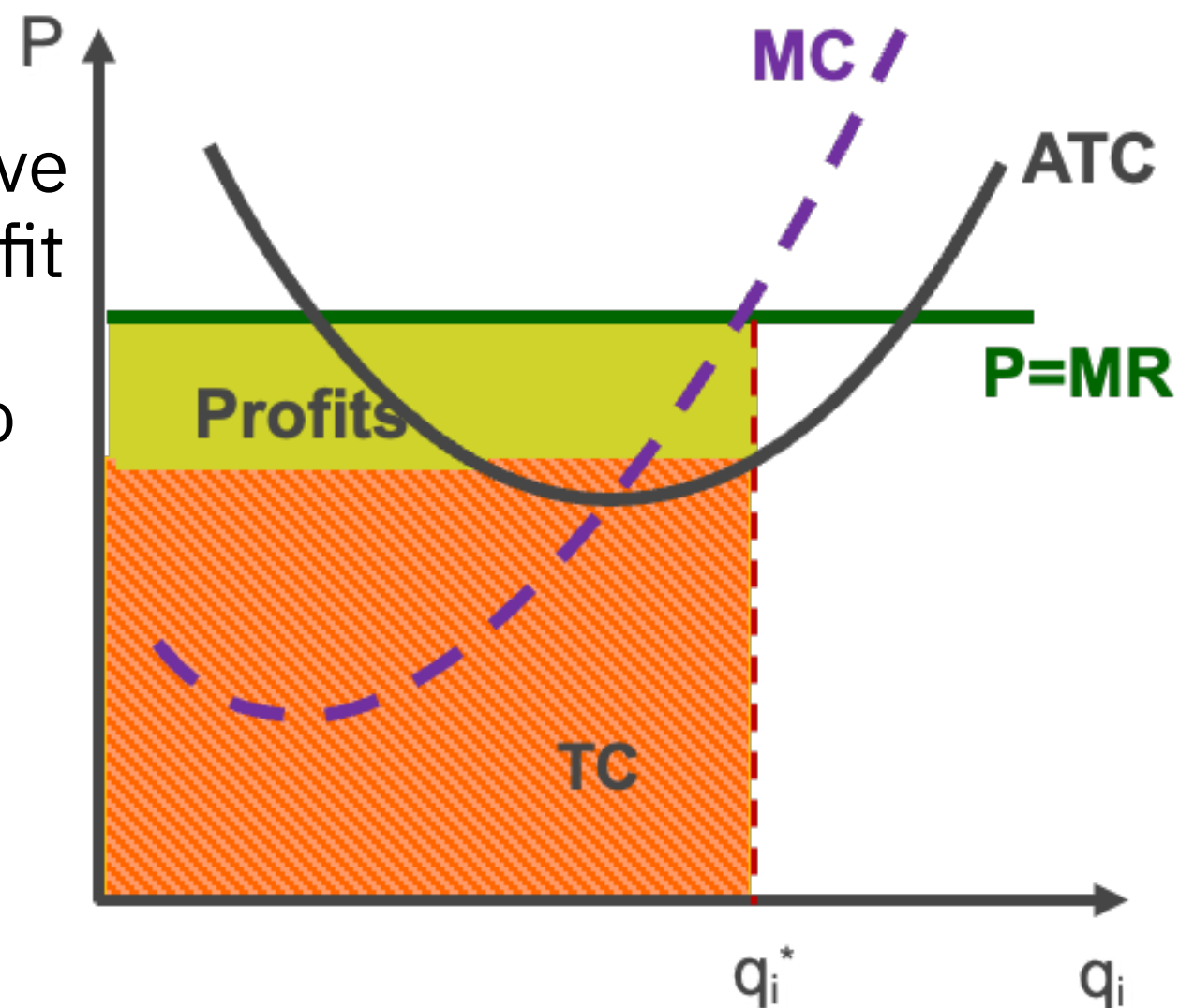
Markets and Competition

Perfect competition

In the short run:

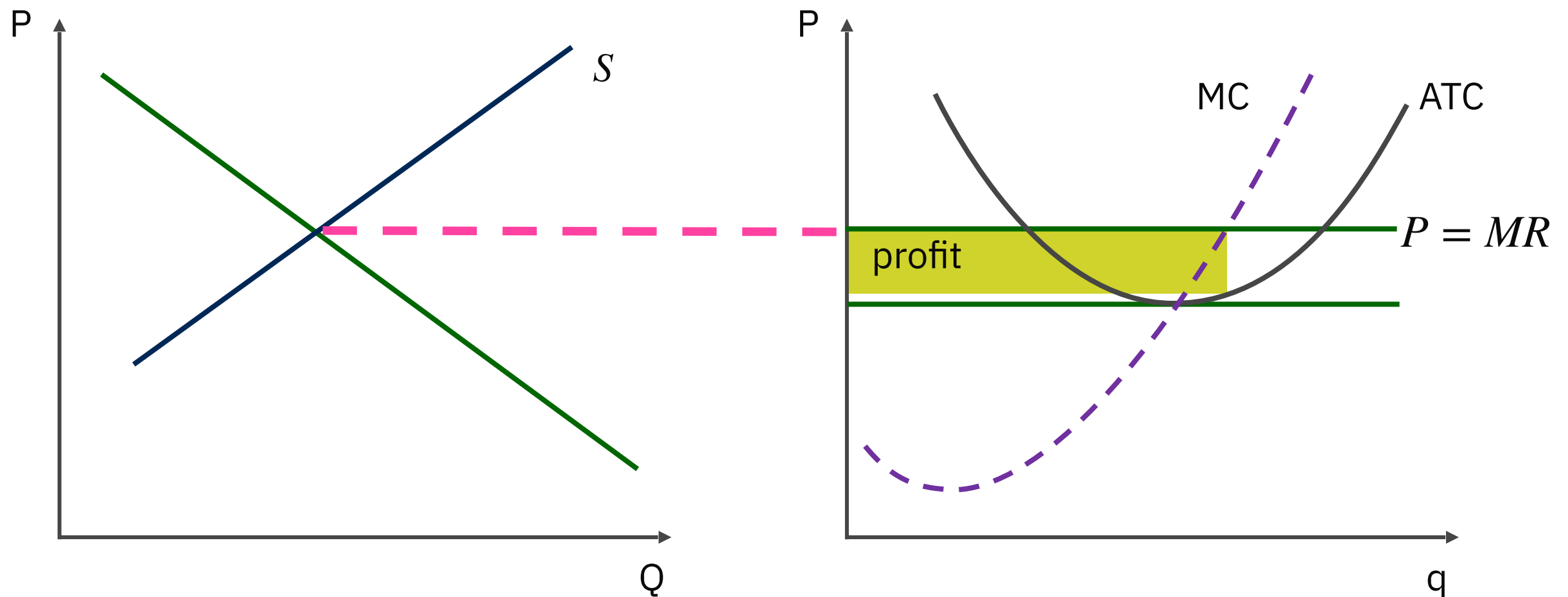
- Possible for firms in competitive markets to earn economic profit
- Profit encourages new firms to enter market
- Market supply shifts to right
- If profits are negative, same story, but in reverse

Let's illustrate this



Markets and Competition

Perfect competition



Markets and Competition

Perfect competition

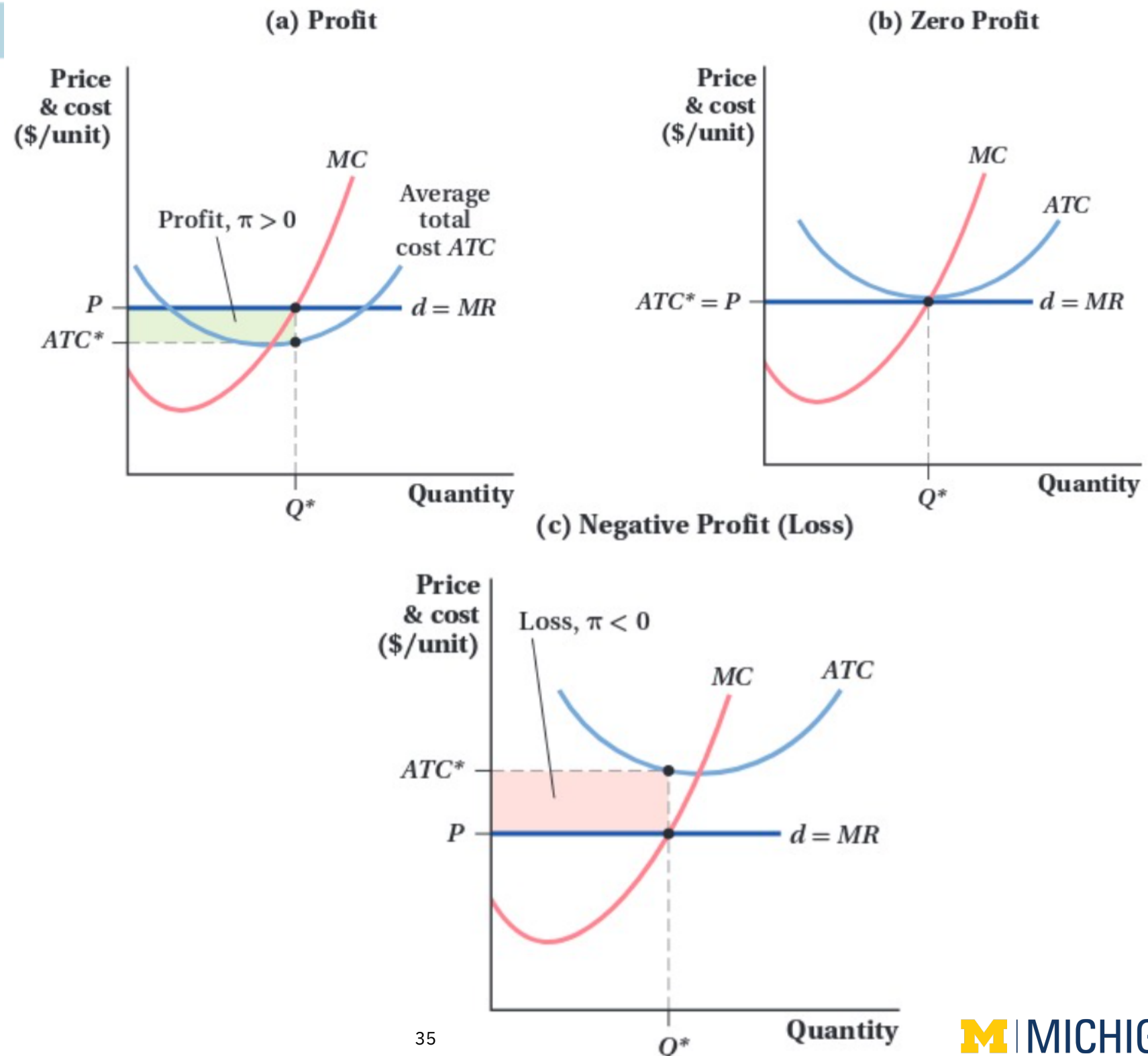
Figure 8.4 Measuring Profit

Given a firm's marginal cost curve MC , average total cost curve ATC , and market price P , the firm's profit can be measured by the area of the rectangle with length Q^* and height $(P - ATC^*)$.

(a) A firm facing a market price above its average total cost curve at Q^* will earn a positive economic profit, $\pi > 0$.

(b) A firm facing a market price equal to its average total cost at Q^* earns zero economic profit.

(c) A firm with average total cost above the market price, at Q^* , will earn negative economic profit (loss), $\pi < 0$.



Markets and Competition

Perfect competition

When there is a short-run view, there is also a long-run view...

Def. Long-run equilibrium is where firms consider only variable costs in making their production decisions

A practical consequence of this definition is that firms view fixed costs as sunk, hence the only thing that characterizes any long-run equilibrium is zero profits

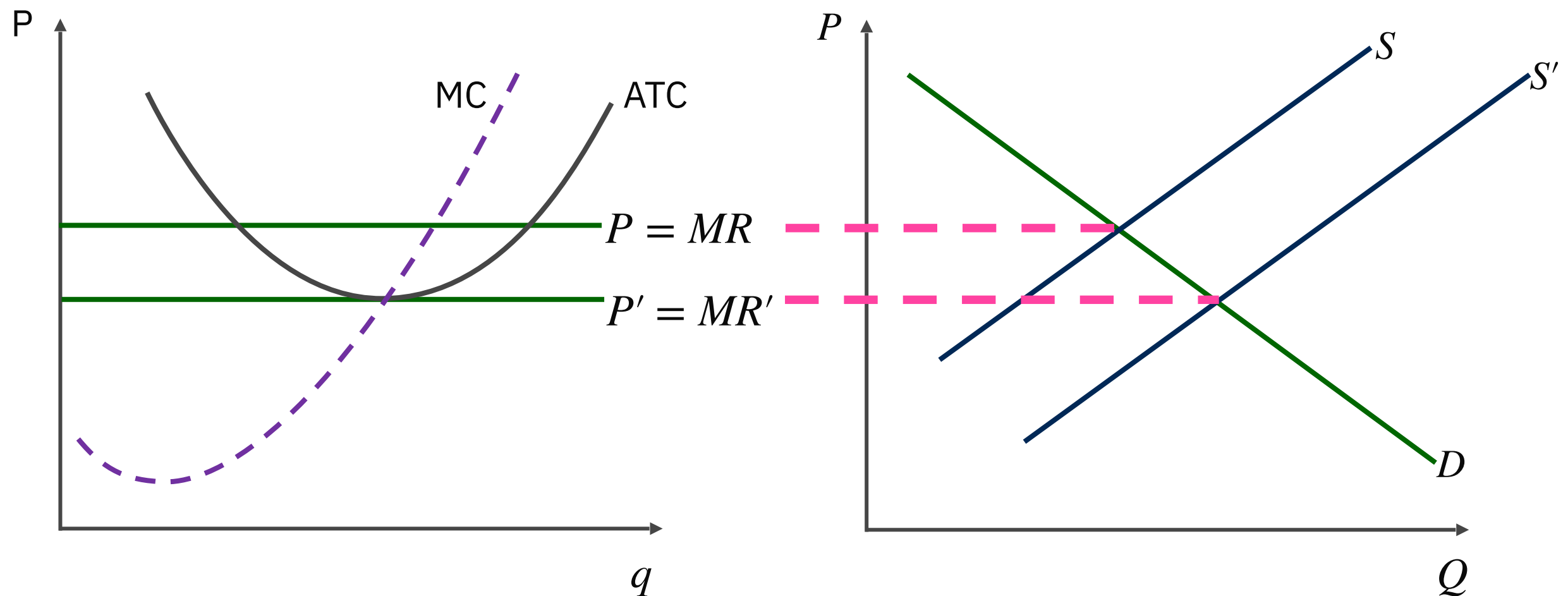
Why?

Because if profits are not zero, there will continue to be free entry and exit, and that is a function of fixed costs, too

Markets and Competition

Perfect competition

In the long-run equilibrium, what must the market price be?



Markets and Competition

Perfect competition

To recap...

Perfect competition describes a market characterized by large numbers of sellers and buyers, each taking the market price as given in making their decisions

Firms set profit-maximizing quantity according to $P = MR = MC$

Markets and Competition

Perfect competition

Short-run (perfectly) competitive equilibrium: economic profits can be positive or negative or zero, there is free entry and exit, and shutdown occurs when the market price cannot cover variable costs

Long-run (perfectly) competitive equilibrium: all firms earn zero economic profits, due to free exit and entry going from short-run to long-run

Markets and Competition

Perfect competition

Example. Fred's Hats, a company involved in the production of University of Michigan Rose Bowl hats, has total cost and marginal cost functions as follows:

$$TC(Q) = q^2 + 2q + 49 ,$$
$$MC(Q) = 2q + 2 ,$$

where q is the quantity of hats produced by Fred's Hats and marginal cost is measured in dollars per unit. Fred's Hats operates in a perfectly competitive market, where the market price for these hats is currently \$20 each.

Calculate the short-run profit maximizing quantity of hats that Fred's Hats should produce, and the profit that they make. Then, state whether the market is in a long-run equilibrium.

Markets and Competition

Perfect competition

Example (continued). Suppose market demand for these hats is given by $Q_D = 220 - 5P_D$ (or $P_D = 44 - 0.2Q_D$), and the cost structure for Fred's Hats remains the same as those on the previous slide.

Calculate the long-run equilibrium market price.

Markets and Competition

Perfect competition

Why do I care about knowing the market structure, i.e. how many competitors there are?

Economists' answer: market structure has direct implications for how well-off people and firms are

Nomenclature: market structure has welfare implications for the society

MORE ON WELFARE ANALYSIS NEXT TIME, AND THAT CONCLUDES PART 1 OF THIS CLASS!

For next time

Come to office hours w/ any questions/concerns

See you next Monday (9am sharp)!